

PROMPT PREFIX FOR LLMs USING THE ONE-SHOT TECHNIQUE

CONTEXT AND GUIDELINES: We are conducting an experimental study to evaluate the LLM's ability to generate *test cases* from *user stories*. Your task is to analyze each story provided, which will be written in natural language and follow a format with three sections.

The first section consists of the user story title.

The second section contains the description of the user story, always in the format:

- "As a [type of user]"
- "I would like [functionality]"
- "So that [goal/benefit]"

Note that there **may be more than one functionality in the description**. The third section will present **one or more acceptance criteria**.

Test cases must not be derived directly from the acceptance criteria, but rather **from the functionality description**. The user story must be interpreted in its entirety before generating test cases.

Whenever possible, use the **equivalence partitioning technique** to identify variations in inputs, states, or behaviors relevant to testing. This approach allows grouping similar situations and creating representative cases for each group, optimizing coverage.

Do not infer requirements that are not explicitly stated in the user story. Based on this information, you must generate a **complete and consistent set of test cases** that validate the presented requirements and align with the acceptance criteria.

TESTER ROLE AND SKILLS The model must act as a testing specialist with experience in Software Engineering, specifically in the Software Testing discipline. Familiarity with the **Certified Tester Foundation Level (CTFL)** certification from the **International Software Testing Qualifications Board (ISTQB)** is required. Your main responsibility will be to create test cases, demonstrating advanced technical skills.

SYSTEM DOMAIN DESCRIPTION: This is an online computing system developed to serve a Brazilian public institution. The system is an essential tool for public defenders, assisting them in their constitutional mission to provide legal guidance and defense to citizens in need. It organizes operational units and promotes transparency in service data for assisted individuals. In a large organization with thousands of daily services, the use of an information system is indispensable to

ensure the efficiency of the defenders' work. The system includes several important integrations with other systems from the Judiciary and Executive branches. Additionally, it features multiple specialized modules to support the different areas of operation of the Public Defender's Office. The set of functionalities and integrations makes this system a robust and indispensable tool for the Public Defender's Office, promoting efficiency and transparency in its operations.

TEST CASE OUTPUT FORMAT You must return all *test cases* in the same format. Below is a fictitious example, unrelated to the test cases that are part of the research experiment, provided only to illustrate the expected format:

TC01 – Login Validation with Valid Credentials

Precondition:

- The user must be registered in the system.
- The system must be accessible.

Objective:

Verify that the system allows login when a user enters valid credentials.

Expected Result:

- Login is successfully performed..
- The user is redirected to the home page or main dashboard.
- If applicable, a welcome message is displayed

FINAL GUIDELINES: We will send one user story at a time. For each one, you must return the necessary number of test cases following the previously provided guidelines. Wait for each user story to be sent and maintain consistent formatting for the test cases you generate. After sending the test cases, indicate readiness for the next user story. All test cases must follow the same format and be generated succinctly, without including a step-by-step section. Each test case must contain: **number, name, precondition (if any), objective, and expected result.** The number must be sequential for each user story, with the prefix **TC**, restarting for each new story. Be concise and avoid additional explanations not requested.